



Blood Test Analysis Dashboard Project Documentation





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Introduction

Healthcare organizations generate a huge amount of medical data every day. Blood tests are among the most important diagnostic tools used to evaluate patient health. This project transforms raw blood test records into interactive dashboards that support healthcare professionals in making informed decisions.

Project Overview

The Blood Test Analysis Dashboard analyzes blood test results for approximately 100,000 patients. It visualizes Hemoglobin, Glucose, Cholesterol, WBC, Platelet Status, Age Distribution, and Gender Distribution using Power BI.

Problem Statement

Healthcare institutions collect massive volumes of laboratory test data daily. Manual analysis is time-consuming and makes it difficult to identify trends and support clinical decisions. This dashboard provides interactive insights for faster decision-making.

Project Objectives

- Analyze blood test records.
- Measure healthcare KPIs.
- Compare indicators by gender.
 - Study age groups.
- Detect patients at risk of diabetes.
- Support healthcare decision-making.



Dataset Description

The dataset contains blood test information for approximately 100,000 patients including Gender, Age, Hemoglobin, Glucose, Cholesterol, WBC, and Platelet Status.

Data Pipeline

Data Ingestion → Data Understanding → Data Cleaning → Data Modeling → DAX Measures → Visualization → Insights Generation.

Dashboard Overview

Main KPIs include Total Patients, Average Hemoglobin, Average Glucose, Average Cholesterol, Average WBC, and Platelet Status.

Dashboard Screens

Add screenshots of:

- Dashboard Overview
 - Age Distribution
 - Gender Analysis
- Hemoglobin Analysis
- Glucose Analysis
- Platelet Status



Key Insights

- 48.4% of patients are pre-diabetic.
- Platelet counts are normal for over 95% of patients.
 - Cholesterol levels are within the healthy range.
 - Male hemoglobin levels are naturally higher than female levels.
- WBC values indicate a generally healthy population.

Challenges

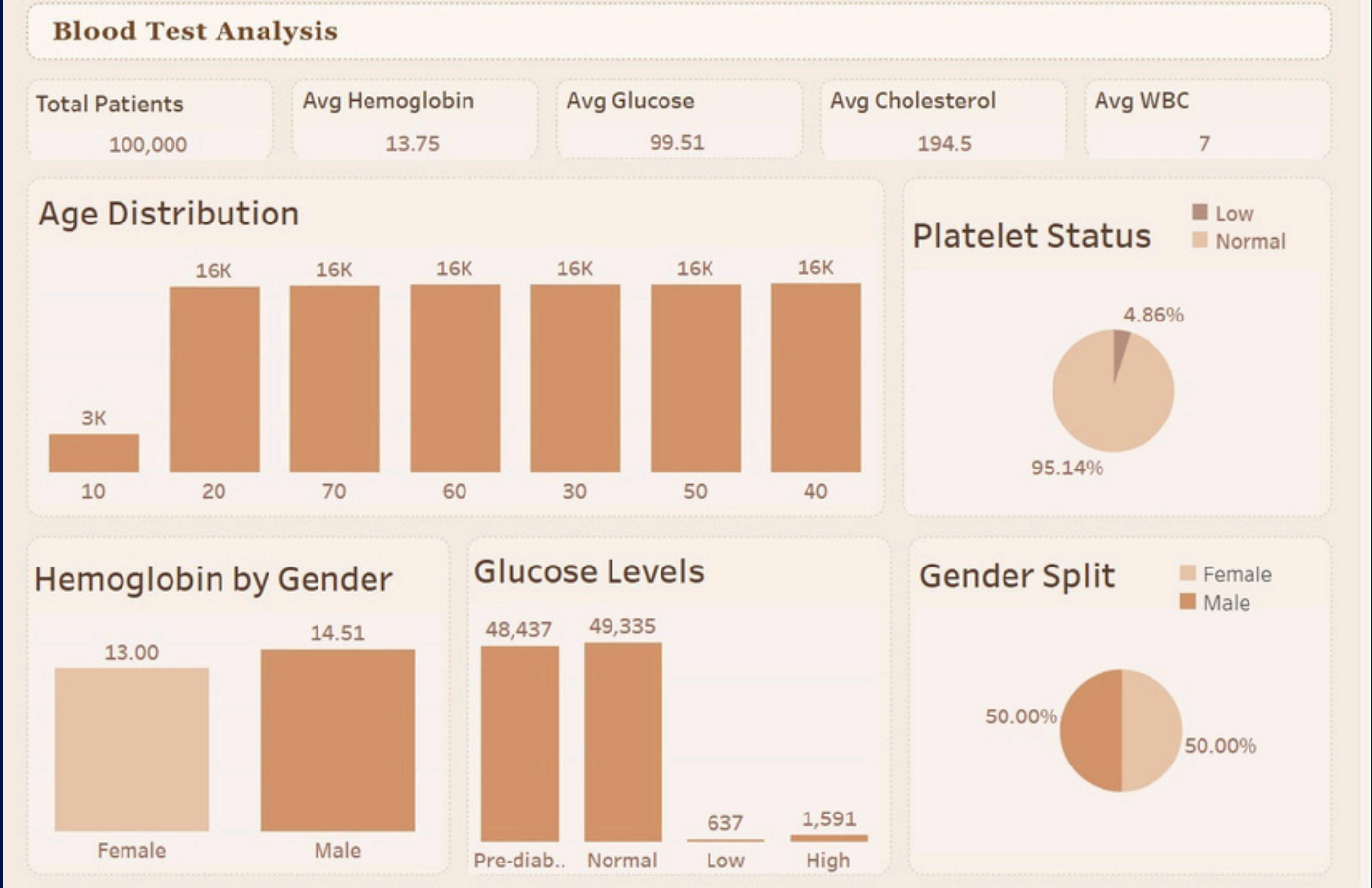
Data cleaning, removing duplicates, handling missing values, designing KPIs, and optimizing dashboard performance.

Future Improvements

Machine learning prediction, real-time dashboards, cloud integration, automated reporting, and predictive healthcare analytics.

Conclusion

The project successfully transformed raw medical data into meaningful healthcare insights using Power BI, SQL, Excel, and DAX.



INTERACTIVE DASHBOARD

Establishing baseline metrics for a 100,000-patient cohort

Total Patients: 100,000

Average Hemoglobin

13.75

Within Standard Physiological Range

Average Glucose

99.51

Within Standard Physiological Range

Average Cholesterol

194.5

Within Standard Physiological Range

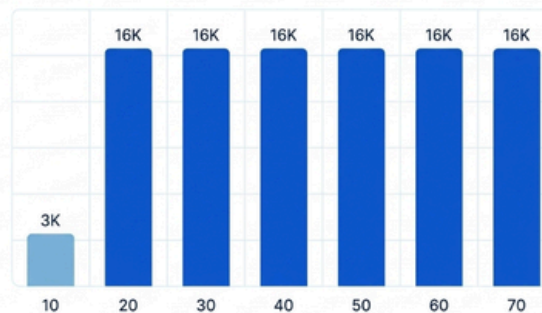
Average WBC

7

Within Standard Physiological Range

NotebookLM

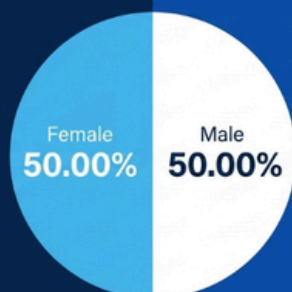
An evenly distributed adult population anchors the clinical data



Uniform adult sampling (16,000 per decade) ensures age-related metabolic findings are not skewed by demographic concentration.

NotebookLM

Perfect gender parity eliminates demographic bias from the clinical findings



With exact gender parity and uniform age distribution, the following metabolic risks represent true population-wide trends, entirely free of sampling bias.

Baseline hemoglobin health with ex

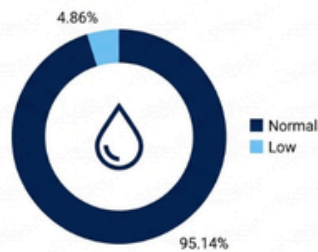
Platelet Sta



Snipping Tool

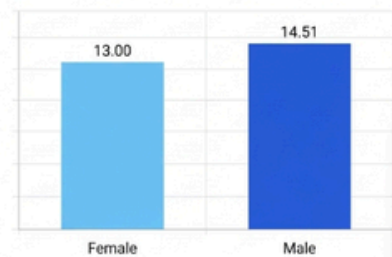
Baseline hematology indicates strong systemic health with expected physiological variances

Platelet Status



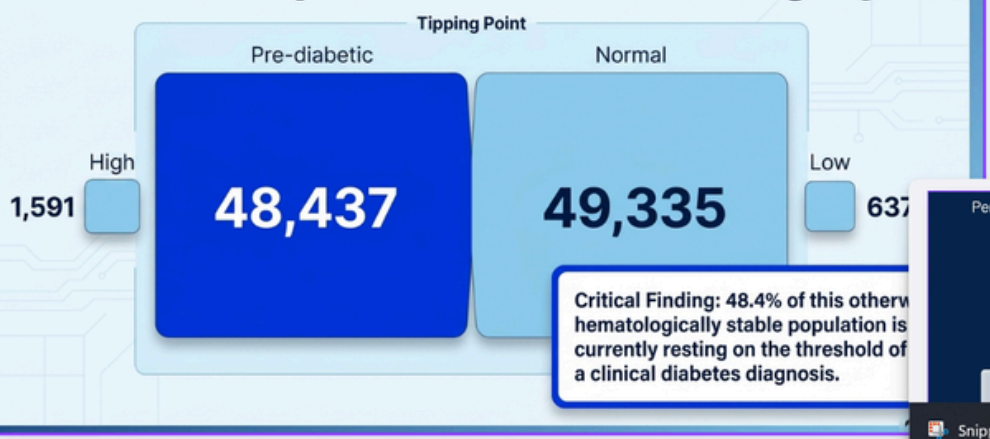
Systemic clotting factors indicate a healthy, stable population baseline.

Hemoglobin by Gender



Standard physiological divergence observed between male and female baselines, aligning with global clinical expectations.

Nearly half of the tested population falls into the pre-diabetic risk category



Stable hematological baselines contrast sharply with looming metabolic risks

Stable Systemic Indicators



Demographics

Perfect 50/50 gender split and uniform adult age distribution (16k/decade).



Hematology

95.14% normal platelet function and expected physiological hemoglobin variance.

Actionable Metabolic Risks



Glucose Anomaly

48,437 patients identified as pre-diabetic, representing an imminent shift in population health.



Cholesterol Watch

Average population cholesterol sits at 194.5, dangerously close to the 200 mg/dL borderline-high threshold.

Recommendation: Reallocate population health resources from general systemic monitoring toward targeted, preventative metabolic and endocrine interventions.